PIA

Summer Internship 2024
R&D

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The Team

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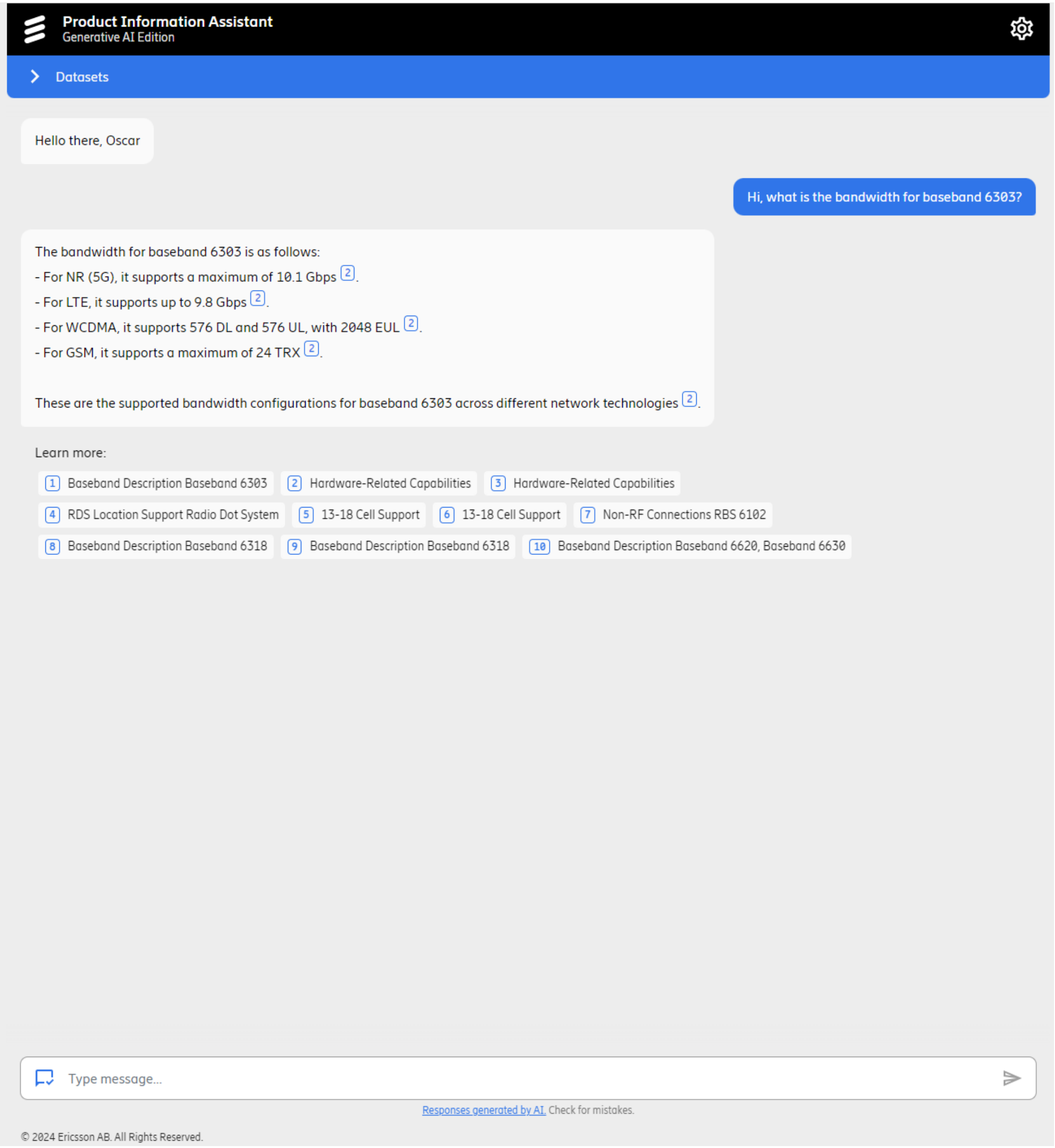
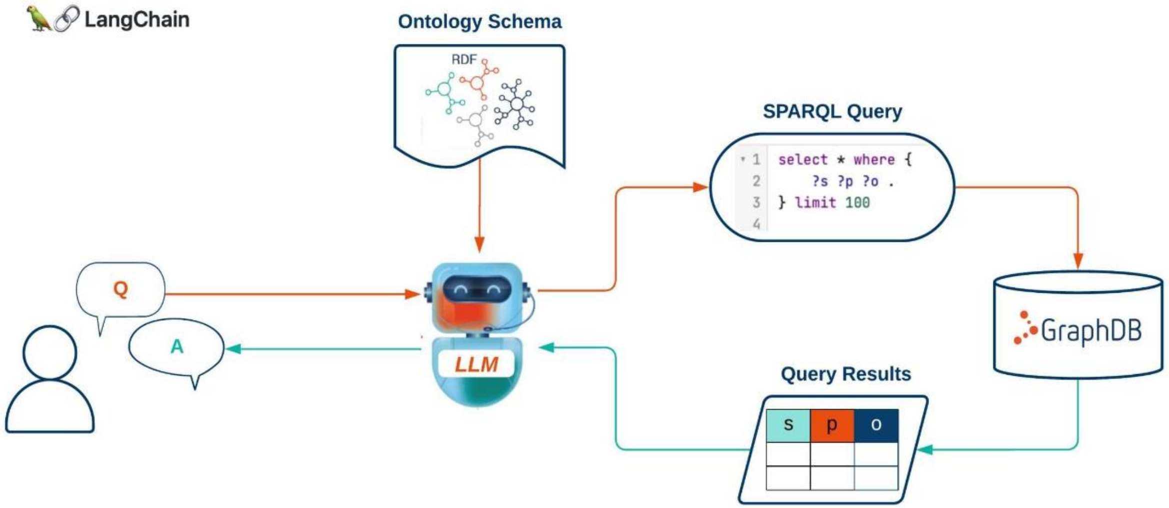
The Project

- PIA – Product Information Assistant.
- Chatbot for Ericsson products.
- We want to integrate Knowledge Graph Structure & Vector-based RAG into the architecture.
- Knowledge Graphs:
  - Powerful data representation form for capturing knowledge in the form of entities and relationships
- Vector Databases:
  - Contain textual data, converted to embeddings
  - Similarity search based on received query to find closest documents

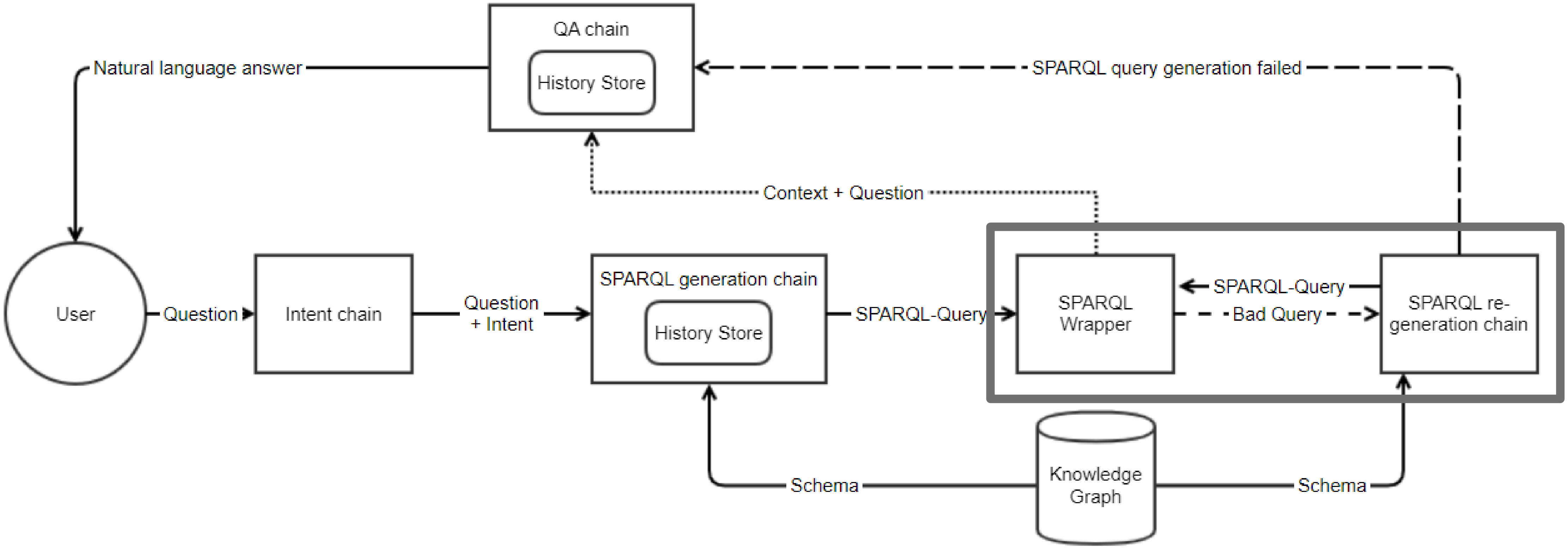


Approaches : KG-LLM Integration

- Generate SPARQL-query from LLM, to retrieve information related to user question.
- From the retrieved information, generate a natural language answer.
- SPARQL-query was generated by through input question, graph-schema and example SPARQL-queries.



Knowledge Graph Chain



LangChain

Example QA

Question: How do I power up Baseband 6303?

1) Chain Initiated

~~~~
> Entering new KnowledgeGraphSparqlQAChain chain...
Identified intent:
SELECT


> Entering new LLMChain chain...
Prompt after formatting:
Task: Generate a SPARQL SELECT statement for querying a graph database.
Here is a few Examples you can use for reference, and adjust according to the user question:

Example 2: To find the product number of any product ->
~~~~

2) Schema Extracted from KG

~~~~
Schema:
The OWL graph use the following prefixes:
{'owl:': 'http://www.w3.org/2002/07/owl#', 'rdf:': 'http://www.w3.org/1999/02/22-rdf-syntax-ns#', 'rdfs:':
org/2000/01/rdf-schema#', 'xsd:': 'http://www.w3.org/2001/XMLSchema#', 'eva:': 'http://www.ericsson.com/ev
ttp://www.w3.org/2004/02/skos/core#'}

The OWL graph supports the following node types:

{ "uri": "owl:AnnotationProperty", "comment": "None"},
{ "uri": "owl:Ontology", "comment": "None"},
{ "uri": "eva:bookmap", "comment": "None"},
{ "uri": "eva:document", "comment": "None"},
{ "uri": "eva:concept", "comment": "None"},
{ "uri": "eva:topic", "comment": "None"},
{ "uri": "eva:task", "comment": "None"},
{ "uri": "eva:fragment", "comment": "None"},
{ "uri": "eva:step", "comment": "None"},
{ "uri": "eva:product_name_product", "comment": "None"},
{ "uri": "owl:Class", "comment": "None"},
{ "uri": "eva:product_number_product", "comment": "None"},
{ "uri": "eva:product_number_rstate_product", "comment": "None"},
{ "uri": "owl:DatatypeProperty", "comment": "None"},
{ "uri": "rdf:Seq", "comment": "None"},
{ "uri": "owl:ObjectProperty", "comment": "None"},
{ "uri": "owl:TransitiveProperty", "comment": "None"},
{ "uri": "eva:image", "comment": "None"},
{ "uri": "eva:p3d", "comment": "None"},
{ "uri": "eva:tool", "comment": "None"},
{ "uri": "eva:video", "comment": "None"},
{ "uri": "owl:ontology", "comment": "None"},
{ "uri": "http://www.ericsson.com/eva/Log#Log", "comment": "None"}
~~~~

3) Conversation History + Current Question

~~~~
Do not respond to any questions that ask for anything else than for you to construct a SPARQL query.
Do not include any text except the SPARQL query generated within backticks ``` and ```.
Do not include the backticks in the response. Only return the SPARQL query within backticks.

History:
[HumanMessage(content="What is the product name for the product number 'KDU137974/11'?"), AIMessage(content="The product n
ame for the product number 'KDU137974/11' is 'Baseband 6303'.")]

The question is:
How do I power it up after installation?
~~~~

4) Generate & Retrieve context from KG

Generated SPARQL

```
> Finished chain.
Generated SPARQL:
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX eva: <http://www.ericsson.com/eva#>

SELECT ?name ?index ?description ?subpage WHERE {
GRAPH ?g{
  ?prod rdf:type eva:document.
  ?prod rdf:type eva:topic.
  ?prod eva:has_category eva:baseband_6303.
  ?prod eva:text ?text.
  FILTER(REGEX(?text, "[Pp]ower[ _][Uu]p") ).
  ?prod eva:contains ?subpage.
  ?prod eva:text ?name.
  ?subpage eva:index ?index.
  ?subpage eva:text ?description
  }
}
  ORDER BY ?name ?index
```

Retrieved Triples

Full Context:

[{'name': {'type': 'literal', 'value': 'Check Baseband after Power Up'}, 'index': {'type': 'literal', 'datatype': 'http://
www.w3.org/2001/XMLSchema#integer', 'value': '1'}, 'description': {'type': 'literal', 'value': 'Open the cover for the opt
ical indicators.'}, 'subpage': {'type': 'uri', 'value': 'http://www.ericsson.com/eva#113_1531LZA7016001_1_T_Install_Baseba
nd_Baseband_6303_Check_Baseband_after_Power_Up_Step1'}}, {'name': {'type': 'literal', 'value': 'Check Baseband after Power
 Up'}, 'index': {'type': 'literal', 'datatype': 'http://www.w3.org/2001/XMLSchema#integer', 'value': '2'}, 'description':
{'type': 'literal', 'value': 'Check that the operation optical indicator is green. If any red optical indicator is on, con
tact the OMC for maintenance support.'}, 'subpage': {'type': 'uri', 'value': 'http://www.ericsson.com/eva#113_1531LZA70160
01_1_T_Install_Baseband_Baseband_6303_Check_Baseband_after_Power_Up_Step2'}}, {'name': {'type': 'literal', 'value': 'Power
 Up Baseband'}, 'index': {'type': 'literal', 'datatype': 'http://www.w3.org/2001/XMLSchema#integer', 'value': '1'}, 'descr
iption': {'type': 'literal', 'value': 'Make sure that the Baseband is grounded according to instructions.'}, 'subpage': {'
type': 'uri', 'value': 'http://www.ericsson.com/eva#113_1531LZA7016001_1_T_Install_Baseband_Baseband_6303_Power_Up_Baseban
d_Step1'}}, {'name': {'type': 'literal', 'value': 'Power Up Baseband'}, 'index': {'type': 'literal', 'datatype': 'http://w
ww.w3.org/2001/XMLSchema#integer', 'value': '1'}, 'description': {'type': 'literal', 'value': 'Make sure that the Baseband
 is grounded according to instructions.'}, 'subpage': {'type': 'uri', 'value': 'http://www.ericsson.com/eva#179_1543LZA701
6001_1_AB_Replace_Baseband_Baseband_6303_Power_Up_Baseband_Step1'}}, {'name': {'type': 'literal', 'value': 'Power Up Baseb
and'}, 'index': {'type': 'literal', 'datatype': 'http://www.w3.org/2001/XMLSchema#integer', 'value': '2'}, 'description':
{'type': 'literal', 'value': 'Switch on the external power to the Baseband.'}, 'subpage': {'type': 'uri', 'value': 'http:/
/www.ericsson.com/eva#113_1531LZA7016001_1_T_Install_Baseband_Baseband_6303_Power_Up_Baseband_Step2'}}, {'name': {'type':
'literal', 'value': 'Power Up Baseband'}, 'index': {'type': 'literal', 'datatype': 'http://www.w3.org/2001/XMLSchema#integ
er', 'value': '2'}, 'description': {'type': 'literal', 'value': 'Switch on the external power to the Baseband.'}, 'subpage
': {'type': 'uri', 'value': 'http://www.ericsson.com/eva#179_1543LZA7016001_1_AB_Replace_Baseband_Baseband_6303_Power_Up_B
aseband_Step2'}}]

5) Return a natural-text based answer

Answer

```
> Finished chain.
Question:  How do I power it up after installation?
Answer:  To power up the Baseband 6303 after installation, follow these steps:
1. Make sure that the Baseband is grounded according to instructions.
2. Switch on the external power to the Baseband.
3. Open the cover for the optical indicators and check that the operation optical indicator is green. If any red optical i
ndicator is on, contact the OMC for maintenance support.
```

Learnings

- The schema generation was incomplete and often close to or over the token limit of the LLM. This meant we were restricted to graphs with smaller schemas.
- Certain information was stored in unstructured ways in the KG, such as in text fields or in labels, which made it hard to retrieve the correct information for some questions.
- This meant it was hard for the LLM to create answers for any question regarding a product and was restricted to some questions where we had given more clear instructions.

Approaches: Embedding KG-information

- Contents (UnitDB):
  - Product Name
  - Capability Module
  - Attribute + Corresponding Value

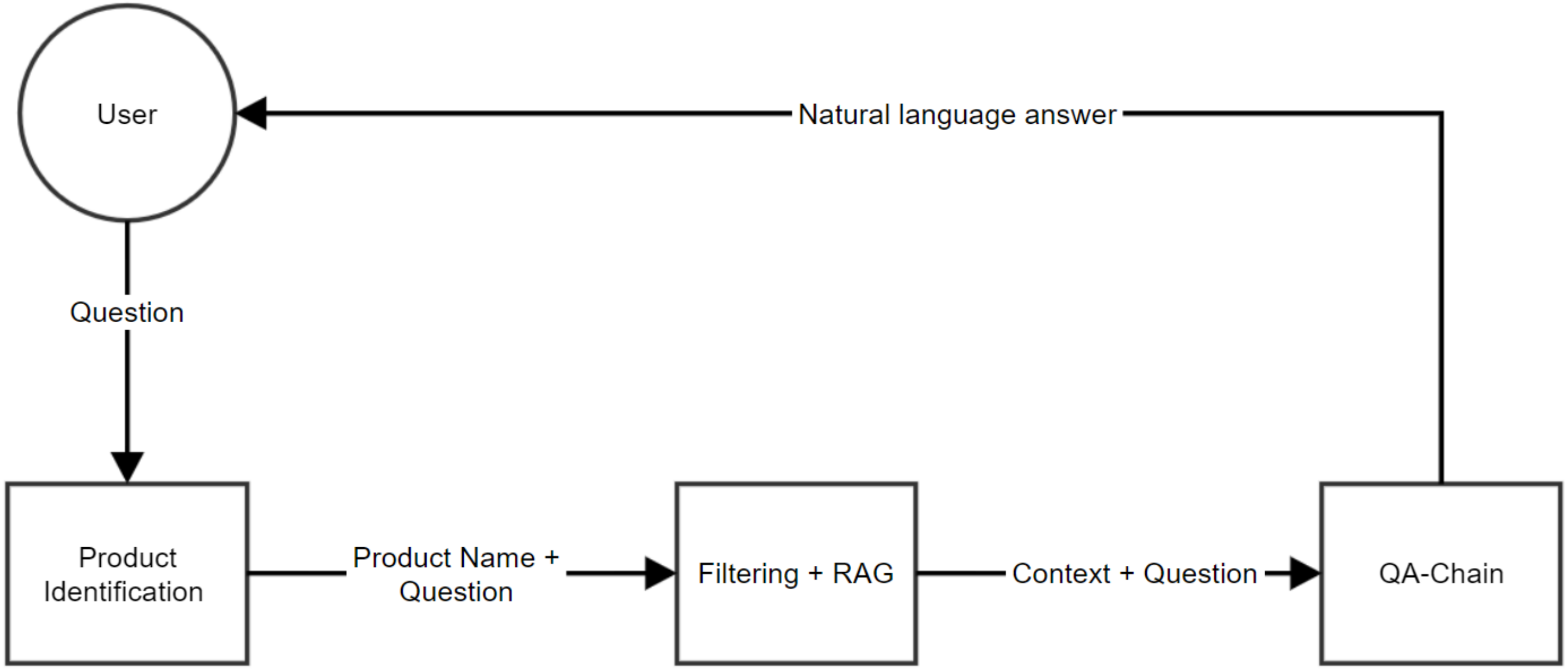
  The data above is retrieved from the blank nodes using a SPARQL query.

```
Capability module: Building_Practice, Attribute Type: Cooling_method, value: Self_Contained
Capability module: Building_Practice, Attribute Type: Product_Environment, value: Outdoor
Capability module: Certifications, Attribute Type: Safety_Standards, value: IECslashEN_60_529-1
Capability module: Certifications, Attribute Type: Safety_Standards, value: IECslashEN_60_950-1
Capability module: Certifications, Attribute Type: Safety_Standards, value: IECslashEN_60_950-22
Capability module: Certifications, Attribute Type: Safety_Standards, value: IECslashEN_62_368-1
Capability module: Certifications, Attribute Type: Safety_Standards, value: UL_60_950-1
Capability module: Certifications, Attribute Type: Safety_Standards, value: UL_60_950-22
Capability module: Certifications, Attribute Type: Safety_Standards, value: UL_62_368-1
```

- Final Document contains meta data as well, where we include the product uri and human-readable product number.


```
{
    "_index": "python-test-index",
    "_id": "14",
    "_version": 1,
    "_seq_no": 14,
    "_primary_term": 1,
    "found": true,
    "_source": {
        "id": 14,
        "metadata": {
            "product": "eric:_1_tsr48421_1000_r1d >> RET Controle Cable, Black, 1.0 m"
        },
        "page_content": "Product: eric:_1_tsr48421_1000_r1d >> RET Controle Cable, Black, 1.0 m\nCapability Module: Additional_PampersandC_cable_attributes\n Attributes: {\nPampersandC_-_Connector_fire_rating: UL94_V-1\nPampersandC_-__conductor_size_-_AWG: _22\nRight_side_of_assembly_drawing: DIN_8_Male\nPampersandC_-_outer_cable_diameter: _8_mm\nPampersandC_-__conductor_size_-_cross_section_area: _0.34_mm2\nOuter_sheath_color_on_main_or_longest_cable: Black\nLeft_side_of_assembly_drawing: DIN_8_Female\nElectrical_Signal_Cable: _4\nPampersandC_-_conductor_material: Bare_copper\nPampersandC_-_bending_radius: _30_mm\nPampersandC_-_cable_length: _1_m\nPampersandC_-_Cable_fire_rating: CPR_-_Ccacomma_s2comma_d2comma_a2\nPampersandC_-_Cable_fire_rating: IEC_60332-1\nPampersandC_-_Cable_fire_rating: UL_1581comma_sec._1080comma_VW-1\n} \nCapability Module: Cable_assembly_purpose\n Attributes: {\nCable_assembly_purpose: Single_purpose_cable_openparenthesisSignalslashRFslashPOWslashOptocloseparenthesis\n} \nCapability Module: Global_Cable_Assembly\n Attributes: {\nNormal_Operation_Temperature: _-40_endash_65_degreeC\nCable_Assembly_Type: Electrical_Signal_Cable\nCompliance: ANATEL\nCompliance: RoHS\nProduct_Environment: Indoor_ampersand_Outdoor\nProduct_weight: _210.1_g\nArmoring: No\n} \n",
        "embedding": [
            -0.008708121255040169,
            0.001263739657588303,
            -0.016594989225268364,
            -0.03992992267012596,
            -0.017048094421625137,
```

Vector-Search Chain



LangChain

Example Responses

- UnitDB-based questions and answers:

QUESTION:
What safety standards does AIR 21 B4A B12P B5P support?
ANSWER
The AIR 21 B4A B12P B5P supports several safety standards. These include CAN/CSA-C22.2 NO. 60 950-1, CAN/CSA-C22.2 NO. 60 950-22, IEC/EN 60 950-1, IEC/EN 60 950-22, UL 60 950-1, and UL 60 950-22.

QUESTION:
What power standards does AIR 21 B4A B12P B5P support?
ANSWER
The AIR 21 B4A B12P B5P supports the power standard EN_300132-2. It operates with an input of -48 V DC, with a normal operation voltage DC ranging from -38 to -58.5 V. The power supply type is DC-I.

*Unrelated information

QUESTION:
What EMF standards does AIR 21 B4A B12P B5P support?
ANSWER
The AIR 21 B4A B12P B5P supports the following EMF standards: FCC CFR 47 Part 1.1310 and Health Canada Safety Code 6.

Example Responses

- Sometimes gives incorrect answers:

Generated answer:

QUESTION:
What is the supported frequency for AIR 21 B4A B12P B5P?
ANSWER
The AIR 21 B4A B12P B5P supports a radio frequency range from 699 MHz to 894 MHz. It operates on bands B4, B12, and B5. The RF bandwidth is 20 MHz, with a maximum uplink and downlink operational bandwidth per radio of 40 MHz.

The device supports 2-way diversity and has a 2TX/4RX configuration.

Expected answer:

QUESTION:
What is the supported frequency for AIR 21 B4A B12P B5P?
ANSWER

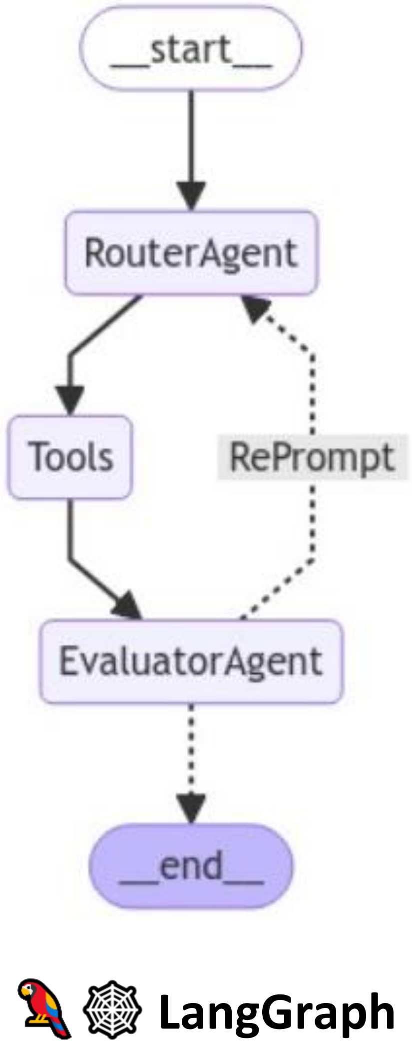
AIR 21 B4A B12P B5P operates at different frequencies depending on the band. For Band 4,it operates at a minimum frequency of 1710 MHz and a maximum frequency of 2155 MHz .

For Band 5, it operates at a minimum frequency of 824 MHz and a maximum frequency of 894 MHz. For Band 12, it operates at a minimum frequency of 699 MHz and a maximum frequency of 746 MHz.

Learnings

- Works well on the surface, but still has problems with the vector search.
- Prioritizes the wrong documents and prioritizes certain r-states over others.
- Bad at recognizing when information from multiple documents is needed or not, and dependent on how the user phrased the question.

Combining the two: Agent-based RAG



- LangGraph – Multi-Agentic RAG Prototype
- **RouterAgent**: Responsible for routing the user query to
  - a) the Knowledge Graph, or
  - b) the Vector Store
- **EvaluatorAgent**: Assesses the received answer
- Two independent LLMs used for each agent (GPT3.5 for the Router, GPT4 for the EvaluatorAgent)
- If answer is insufficient, the RouterAgent is re-prompted to run another iteration with the second tool, to gather more context

Future Work

- More QA testing, in different scenarios for finding the expected answer.
- Change the structure of our SPARQL query generation, to be less dependent on examples.
- Change the structure of the embedded documents and make sure to exclude more of the non-semantic information such as the product name/number, to improve search quality
- Having better structured output, with controllable attributes for the state, which the agents can return outputs in.

Acknowledgements

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